The Great Bitwise Bakeoff project report

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Abstract

We present a system that evolves dessert recipes from 1742 Italian dolci recipes. The system has no hand-written ingredient taxonomy. Its only culinary knowledge is a co-occurrence network, partitioned into Leiden communities. A genetic algorithm evolves ingredient sets using community-aware crossover and affinity-weighted mutation. Three fitness profiles trade off network-based value against novelty. A large language model (LLM) judges the resulting candidates and writes cooking instructions for the finalist, which are presented in a small cookbook. We discuss what co-occurrence can and cannot capture, and how the systems creativity might be evaluated.

Introduction

Recipes are a rich domain for computational creativity. They combine discrete, interpretable components (ingredients, quantities and methods) into artifacts that can be judged on both novelty and value. Dessert recipes are a nice subset as they follow well-understood structural conventions but allow experimentation with unexpected ingredient combinations. The PIERRE (Morris et al. 2012) system demonstrated that an evolutionary approach, seeded by a large corpus of existing recipes, can produce novel recipes that remain recognizable as baked goods. This report describes a similar system for generating dessert recipes, along with a small cookbook that presents its output, since presentation partly shapes how a recipes value is perceived.

The Approach

Inspiring set and preprocessing

Our inspiring set comes from *GialloZafferano*, a popular Italian cooking website, through a publicly available recipe dataset (KanuSaru ; Mondadori Group). The choice of Italian cuisine was motivated by the availability of recipes and by a completely impartial suggestion from the Italian member of the group. We selected the *Dolci* (desserts) category, which includes cookies and cakes alongside other desserts. This gave us a broader set of combinations from which to generate new recipes.

We normalized ingredient names by standardizing case, whitespace and apostrophes, then converted quantities to grams where possible. Explicit mass units were converted directly, for volumes, we approximated $1\,\mathrm{ml}$ as $1\,\mathrm{g}$. Ingredient counts (e.g., eggs) were estimated using a small conversion table. Unspecified quantities such as *q.b.* (*quanto basta*, "as needed") were omitted from mass calculations. For each recipe r, we calculated the relative mass of ingredient i as

$$q_{ri} = \frac{m_{ri}}{\sum_j m_{rj}}, \tag{1}$$

where m_{ri} is the estimated mass. We collected these fractions across recipes whose quantities were fully resolved, obtaining an empirical quantity distribution for each ingredient. The generator can sample from these observed proportions when assembling recipes.

From recipes to an ingredient network

We first constructed a bipartite graph connecting recipes to their ingredients. An edge exists when an ingredient appears in a recipe. The resulting graph has **2,388 nodes** (1,742 recipes and 646 distinct ingredients) and **16,254 edges**. If A denotes its binary recipe–ingredient incidence matrix, the ingredient co-occurrence network is obtained from

$$W = A^{\mathsf{T}}A, \qquad W_{ii} = 0. \tag{2}$$

For distinct ingredients i and j, the edge weight W_{ij} counts how many recipes contain both. This projection has **646 nodes** and **13,873 weighted edges**.

Figure 1 shows both networks. Node sizes reflect degree: in the bipartite graph, this is the number of ingredients in a recipe or the number of recipes containing an ingredient, in the projected graph, it is the number of distinct co-occurring ingredients. Black nodes in the bipartite graph are recipes. Ingredient nodes use the same community colors in both visualizations.

Emergent ingredient communities

We applied weighted Leiden community detection (Traag, Waltman, and van Eck 2019) to the ingredient network. To obtain groups of manageable size, we modified the procedure to split communities larger than 35 ingredients by rerunning Leiden on their subgraphs at increasing resolutions. Communities smaller than eight ingredients were merged with strongly connected neighboring groups when

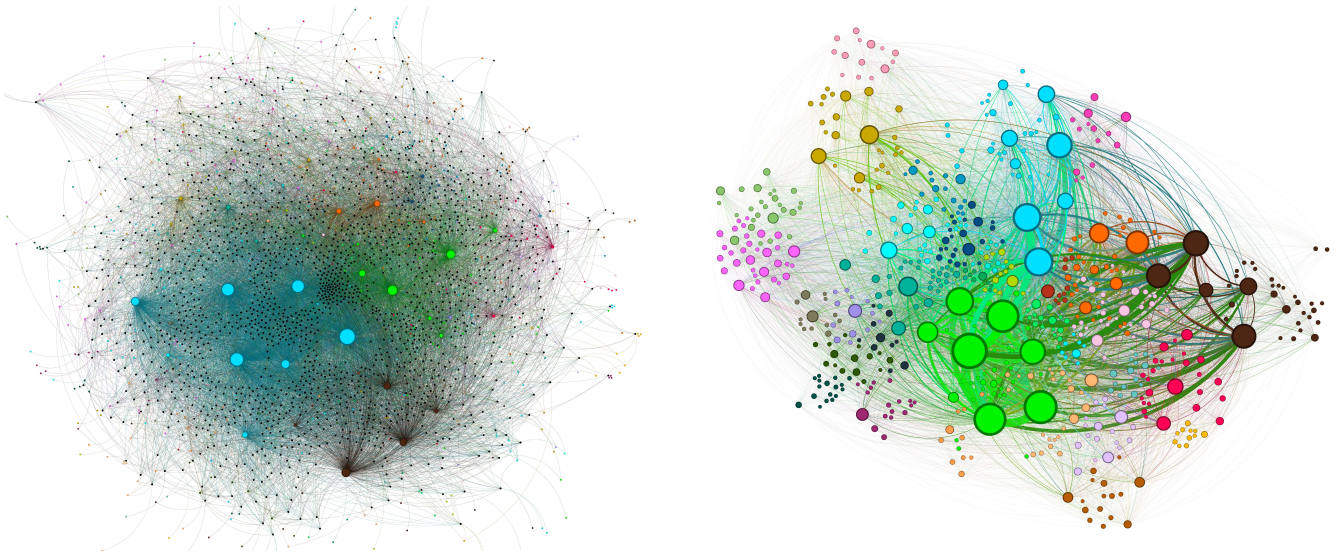


Figure 1: Recipe–ingredient bipartite graph (left; 2,388 nodes, 16,254 edges) and ingredient co-occurrence network (right; 646 nodes, 13,873 edges). Node size encodes degree; colors identify Leiden ingredient communities. In the bipartite graph, black nodes represent recipes and colored nodes represent ingredients. Both networks were visualized using Gephi.

the merged group would not exceed 35 ingredients. These size limits are targets, since some groups cannot be split or merged appropriately.

We deliberately provided no ingredient taxonomy or predefined compatibility rules: there are no labels such as "sweet", "bitter" or "binding agent". Ingredient relationships emerge entirely from their co-occurrence in the source recipes. For example, one community groups several forms of coffee (*caffè*, *caffè solubile*, *caffè in polvere*, and *chicchi di caffè*) alongside *mascarpone* and *savoiardi*, suggesting familiar combinations such as tiramisù. However, the same community also contains less related ingredients, such as *zucchine* (zucchini). These communities therefore reflect patterns in the inspiring set rather than guaranteed flavor compatibility, providing the evolutionary algorithm with a basis for exploring both familiar and unexpected ingredient combinations.

Evolutionary algorithm

A genotype is a set of 5-10 ingredient names. Quantities are added afterwards. Because raw counts are dominated by staples (basic ingredients like sugar, flour, eggs and butter that appear in almost every recipe), the GA uses cosine affinity $a_{ij} = \frac{W_{ij}}{\sqrt{n_i n_j}}$ where n_i is the number of recipes containing i. The GA starts from 60 real recipes and uses tournament selection (size 3) with an elite of 3. Crossover gives the child one ingredient from each community that both parents share, then fills it from the parents remaining ingredients up to 6-9 ingredients. This keeps the "skeleton" the parents agree on. Mutation ($p_m = 0.45$) replaces, adds or removes an ingredient. Replacements normally come from the same community again, but with a creative rate of 0.12 they come from a different one. This is our explicit source of surprise. Candidates are sampled in proportion to their affinity with the rest of the recipe. Duplicate children are rejected. A run stops after 100 generations, or after 15 generations without improvement.

Fitness is a weighted sum of six terms (see Table 1) where each term x is $x \in [0, 1]$ minus a penalty:

- structure: the share of ingredients in the largest connected component.
- compatibility: mean pairwise affinity a_{ij} between the recipe's ingredients.
- novelty: $1 - \max_S J(R, S)$ where J is the Jaccard similarity between recipe R and source recipe S, as in Simple PIERRE (Morris et al. 2012).
- diversity: entropy of the communities in the recipe.
- length: closeness to the 6-9 ingredient range.
- quantity support: the share of ingredients with known quantities.
- penalty: mean $\frac{1}{n_i}$ which discourages rare, noisy ingredients.

Rather than picking one weighting, we run 3 profiles that shift the balance from value to novelty.

Quantities are sampled independently from each ingredients observed distribution of q_{ri} and normalized to a 500g batch (the measured ingredients add up to 500 grams).

Curation and presentation

Each profile runs with 5 seeds, and 20 dissimilar candidates per profile are kept. An LLM (gpt-5.6-terra) scores all 60 candidates on potential (could these ingredients and quantities make a credible dessert?) and creativity (is the combination meaningfully unusual?), each from 0 to 10. The judge does not see the fitness or the profile. Candidates with potential ≥ 5 are selected for the cookbook: 5 Promising (ranked

Component	Cons.	Bal.	Exp.
Structure	0.38	0.30	0.24
Compatibility	0.30	0.25	0.18
Novelty	0.10	0.20	0.32
Diversity	0.08	0.10	0.12
Length	0.08	0.08	0.07
Quantity support	0.06	0.07	0.07
Penalty	0.35	0.30	0.25

Table 1: Fitness weights per profile. Cons. is conservative, Bal. is balanced and Exp. is experimental. The penalty weight is subtracted.

Profile	Fitness	Novelty	Compat.	Core
Cons.	0.752	0.39	0.48	85%
Bal.	0.730	0.66	0.20	10%
Exp.	0.769	0.84	0.03	0%
Inspiring set	–	–	–	75%

Table 2: Mean fitness, novelty and compatibility of the 20 candidates per profile (5 seeds each). Core is the share of recipes containing a sugar, a starch, and an egg or fat. Fitness is not comparable across profiles, since the weights differ.

by potential) and 5 Creative (ranked by creativity). For each finalist, the LLM then writes a English language step-by-step cooking instruction set on how to make the recipe. It may not add, remove or change ingredients, and it must list the recipes limitations honestly.

Results & Discussions

Table 2 summarizes the 20 candidates kept per profile. All runs stopped early except two experimental runs, which reached the 100-generation limit.

Value vs. novelty The profiles behave as intended: mean novelty rises from 0.39 to 0.84 while mean compatibility falls from 0.48 to 0.03, and no candidate is identical to a source recipe. Conservative runs give classic bases, such as vanilla shortbread from cream, butter, sugar and flour (potential 8, creativity 2). Experimental runs give combinations such as kiwi, oat flakes and red currants (7, 6) or red currants, nutmeg and hazelnut flour (7, 5).

Fitness vs. judged value For the ten cookbook recipes, the judge's creativity score closely follows graph novelty (Spearman $\rho = 0.85$), but graph fitness says little about judged potential ($\rho = 0.25$), and novelty is inversely related to it ($\rho = -0.54$). With only ten recipes, all with potential ≥ 5, these figures are indicative only. The judge's reservations concern amounts (too much flour, egg or milk, too little sugar) or a missing sweetener or flavoring, never clashing flavors. The limitations listed with the instructions add missing eggs, leavening or setting agents. Co-occurrence knows what goes together but not how much, and sampling each quantity independently sometimes gives ratios no baker would accept.

What communities capture The basic structure of a dessert is not preserved as novelty increases. 75% of source recipes contain a sugar, a starch, and an egg or fat, 85% of conservative candidates keep this core, but only 10% of balanced and none of the experimental ones, which rarely contain eggs (5%) or sugar (35%). Because the fitness function has no notion of an ingredient's role, a higher novelty weight trades this structure away. The structure score was 1.0 for every candidate, so it did not help to separate them. Communities also cannot detect flavor clashes, so the system cannot tell an inspired pairing from a statistical accident.

Scope Because we used all dolci, none of the ten cookbook recipes is a conventional cookie or cake: five are custards or chilled creams, four are baked bars or slices, and one is a fudge-like confection. This widens the creative space, but it leaves the choice of dessert type to the LLM.

Evaluating the creativity of the system

We evaluate creativity in terms of the balance between novelty and value. Novelty is estimated computationally through distance from the inspiring set, while value is approximated through structural connectivity (which in practice did not discriminate), ingredient compatibility, quantity support, and an LLM judgment of culinary potential made without access to fitness or profile. The three fitness profiles provide different trade-offs between these objectives: the conservative profile favors familiar and well-supported combinations, whereas the experimental profile places greater weight on novelty and therefore produces more unusual recipes. The final candidates are also scored by an LLM for both potential and creativity without revealing their evolutionary fitness or profile, giving a second perspective on the outputs. Overall, the outputs suggests creative behavior because it can generate combinations that are different from the source corpus while remaining recognizable as plausible desserts. However, creativity is not guaranteed by novelty alone: some unusual combinations result from weak co-occurrence patterns or unrealistic independently sampled quantities. A stronger future evaluation would therefore include human tasting or expert assessment to judge whether the generated recipes are not only novel, but genuinely valuable and enjoyable.

Use of Generative AI

Our system generated the ingredient sets, quantities, recipe titles and fitness scores. The LLM generated the judge scores and their one-sentence assessments, the English translations, and the step-by-step cooking instructions. For all ten recipes it also listed physical-form changes, such as reducing milk to thicken a chocolate cream, grinding oats into flour, or melting white chocolate as the only available binder, and the recipe's limitations. These limitations point to concrete weaknesses: missing ingredients, such as eggs or leavening (giving a dense texture), a sweetener (giving a

bitter custard) or a setting agent, and unbalanced amounts (too much salt or flour, too little cocoa). The LLM did not discard awkward ingredients, for example, it blended mint into a custard instead of straining it out. None of the recipes used the option of a separate serving component.

The cookbook images were generated with the gpt-image-2 model, and the short dish names on the covers (e.g., "Mint Chocolate Fudge") were written by the LLM. The images make the recipes noticeably more appealing, but they are not faithful to them. The model added garnishes and ingredients that do not appear in the recipes, such as flaked almonds, whole cloves, fresh mint leaves, sea-salt flakes and chocolate chunks, and it showed dense, unsweetened bakes as fluffy, sugar-dusted cakes. The images therefore persuade partly by misrepresenting the recipe, which conflicts with the honest limitations listed on the same page. This illustrates how much presentation shapes the perceived value of a generated recipe, independently of the recipe itself.

Conclusion

Using only a co-occurrence network learned from 1742 Italian desserts, our evolutionary system produces ingredient sets that an LLM judge rates as plausible and sometimes creative. Its main weakness is quantities. Future work could sample quantities jointly, add knowledge of flavor clashes and include the judge in the evolutionary loop.

References

KanuSaru. Italian recipes. Hugging Face Datasets, `https://huggingface.co/datasets/KanuSaru/italian-recipes`.

Mondadori Group. 2026. digital: Publishing brands. `https://www.mondadorigroup.com/our-business/digital-2026`.

Morris, R. G.; Burton, S. H.; Bodily, P. M.; and Ventura, D. 2012. Soup over bean of pure joy: Culinary ruminations of an artificial chef. In *ICCC*.

Traag, V. A.; Waltman, L.; and van Eck, N. J. 2019. From louvain to leiden: guaranteeing well-connected communities. *Scientific Reports* 9:5233.